

2.1

(a.)

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

Sample Mean: $\bar{x}=1$ $\bar{y}=2$

(b.)

$$b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8 \Rightarrow \text{when } x \text{ increases by 1 unit, the value of } y \text{ increases by 0.8 unit.}$$

$$b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 \times 1 = 1.2 \Rightarrow \text{when } x=0, \text{ the value of } y \text{ is 1.2.}$$

(c.)

$$\sum_{i=1}^5 x_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 15$$

$$\sum_{i=1}^5 x_i y_i = 3 \times 4 + 2 \times 2 + 1 \times 3 + (-1) \times 1 + 0 \times 0 = 18$$

$$\sum (x_i - \bar{x})^2 = \sum x_i^2 - N \bar{x}^2 \leftrightarrow 10 = 15 - 5 \times 1^2$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N \bar{x} \bar{y} \leftrightarrow 8 = 18 - 5 \times 1 \times 2$$

(d.)

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2.0	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$S_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / (N - 1) = (2^2 + 0^2 + 1^2 + (-1)^2 + (-2)^2) / (5 - 1) = 2.5$$

$$S_x^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / (N - 1) = (2^2 + 1^2 + 0^2 + (-2)^2 + (-1)^2) / (5 - 1) = 2.5$$

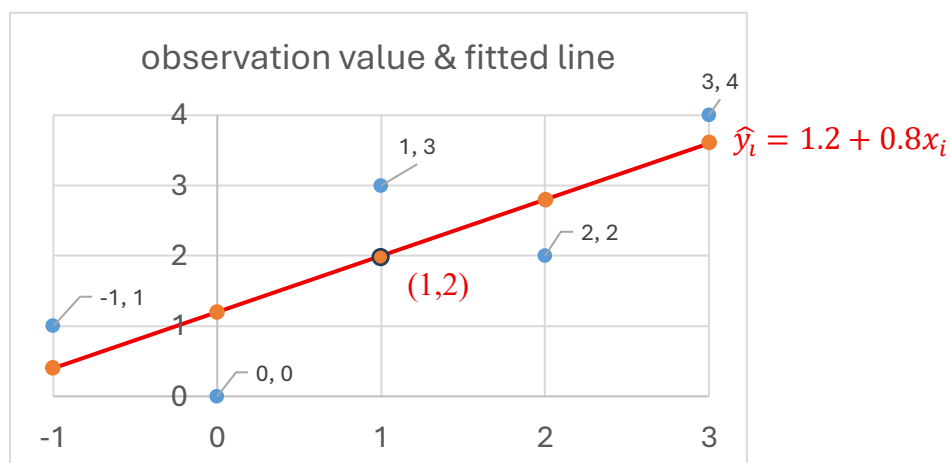
$$S_{xy} = \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) / (N - 1) = 8 / (5 - 1) = 2$$

$$r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{2}{2.5} = 0.8$$

$$CV_x = 100 \left(\frac{S_x}{\bar{x}} \right) = 100 \times \frac{\sqrt{2.5}}{1} \approx 158.11388$$

Median, 50th percentile of x= 1

(e.)



(f.)

The fitted line (Red Line) passes through the point $(\bar{x}, \bar{y}) = (1, 2)$.

(g.)

$$\bar{y} = 2 \quad \bar{y} = b_1 + b_2 \bar{x} \quad \leftrightarrow \quad 2 = 1.2 + 0.8 \times 1$$

(h.)

$$\frac{\sum \hat{y}_i}{N} = \frac{10}{5} = 2$$

(i.)

$$\hat{\sigma}^2 = \frac{\sum (y_i - b_1 - b_2 x_i)^2}{N - 2} = \frac{3.6}{5 - 2} = 1.2$$

(j.)

$$\widehat{Var}(b_2|x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$$

$$se(b_2) = \sqrt{\widehat{Var}(b_2|x)} = \sqrt{0.12} = 0.3464$$

2.22

(a.)

```
Call:
lm(formula = totalscore ~ small, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-191.63  -50.63   -7.12   42.37  324.38

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  918.628     2.536  362.260  <2e-16 ***
small         9.989      4.611   2.166   0.0305 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.36 on 1198 degrees of freedom
Multiple R-squared:  0.003903,    Adjusted R-squared:  0.003071
F-statistic: 4.694 on 1 and 1198 DF,  p-value: 0.03047
```

$$\widehat{totalscore}_i = 918.628 + 9.989 * small_i$$

因為在統計上顯著，故平均而言，若小孩屬小班制，則其學習總分數相較於屬普通班會高出

9.989 分。

(b.)

```
Call:
lm(formula = readscore ~ small, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-76.123 -21.123  -3.356  15.877 192.877

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  434.123     1.044  416.021  < 2e-16 ***
small         5.466      1.897   2.881   0.00403 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 1198 degrees of freedom
Multiple R-squared:  0.006882,    Adjusted R-squared:  0.006053
F-statistic: 8.301 on 1 and 1198 DF,  p-value: 0.004032
```

$$\widehat{readscore}_i = 434.123 + 5.466 * small_i$$

因為在統計上顯著，故平均而言，若小孩屬小班制，則其閱讀分數相較於屬普通班會高出

5.466 分。

```

Call:
lm(formula = mathscore ~ small, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-145.505  -35.028   -6.505   28.495  141.495

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  484.505      1.677 288.884  <2e-16 ***
small         4.522       3.049   1.483   0.138
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.52 on 1198 degrees of freedom
Multiple R-squared:  0.001832,    Adjusted R-squared:  0.0009992
F-statistic: 2.199 on 1 and 1198 DF,  p-value: 0.1383

```

$$\widehat{mathscore}_i = 484.505 + 4.522 * small_i$$

因為在統計上不顯著，所以無法有信心說，若小孩屬小班制，則其數學分數相較於屬普通班會高出 4.522 分的影響。

「因此對於學習總分數的正顯著影響，很可能主要是由對閱讀分數的正向且顯著的影響所驅動。」

(c.)

```

Call:
lm(formula = totalscore ~ aide, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-195.14  -52.14   -6.95   40.95   330.86

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  922.145      2.640 349.250  <2e-16 ***
aide        -1.396       4.437  -0.315   0.753
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.5 on 1198 degrees of freedom
Multiple R-squared:  8.267e-05,    Adjusted R-squared:  -0.000752
F-statistic: 0.09904 on 1 and 1198 DF,  p-value: 0.753

```

$$\widehat{totalscore}_i = 922.145 - 1.396 * aide_i$$

因為在統計上不顯著，所以無法有信心說，若小孩屬有教學助理班級，則其學習總分數相較非屬有教學助理班級來說會有微幅減少 1.396 分的影響。

(d.)

```
Call:
lm(formula = readscore ~ aide, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-77.908 -21.536  -2.908  15.464 191.092

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  435.9084     1.0882  400.585  <2e-16 ***
aide         -0.3719     1.8285   -0.203    0.839
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.29 on 1198 degrees of freedom
Multiple R-squared:  3.453e-05,    Adjusted R-squared:  -0.0008002
F-statistic: 0.04137 on 1 and 1198 DF,  p-value: 0.8389
```

$$\widehat{readscore}_i = 435.9084 - 0.3719 * aide_i$$

因為在統計上不顯著，所以無法有信心說，若小孩屬有教學助理班級，則其閱讀分數相較非屬有教學助理班級來說會有微幅減少 0.3719 分的影響。

```
Call:
lm(formula = mathscore ~ aide, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-147.236 -32.236  -7.212   27.788  140.788

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  486.236     1.744  278.731  <2e-16 ***
aide         -1.024     2.931   -0.349    0.727
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.56 on 1198 degrees of freedom
Multiple R-squared:  0.0001019,    Adjusted R-squared:  -0.0007327
F-statistic: 0.1221 on 1 and 1198 DF,  p-value: 0.7268
```

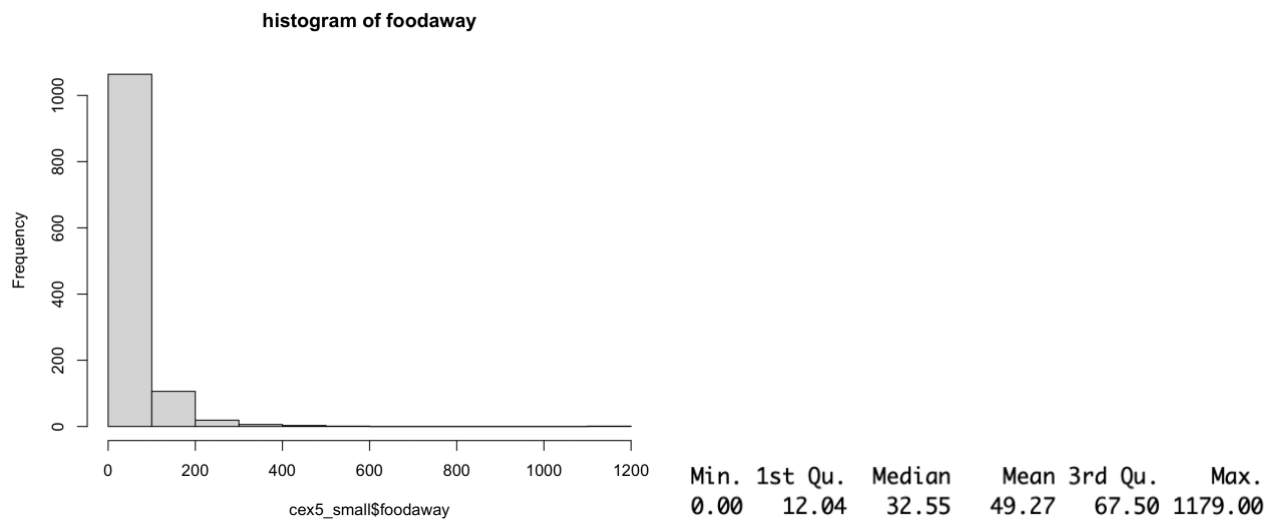
$$\widehat{mathscore}_i = 486.236 - 1.024 * aide_i$$

因為在統計上不顯著，所以無法有信心說，若小孩屬有教學助理班級，則其數學分數相較非屬有教學助理班級來說會有微幅減少 1.024 分的影響。

「因此不論是對於閱讀或數學分數而言，小孩是否處於有教學助理的班級對成績的影響在統計上並沒有顯著意義存在。但他也完美詮釋為何在對於學習總分數影響的不顯著。」

2.25

(a.)



Mean= 49.27

Median=32.55

25th percentile=12.04

75th percentile=67.50

(b.)

advanced=1

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	21.67	48.15	73.15	90.00	1179.00

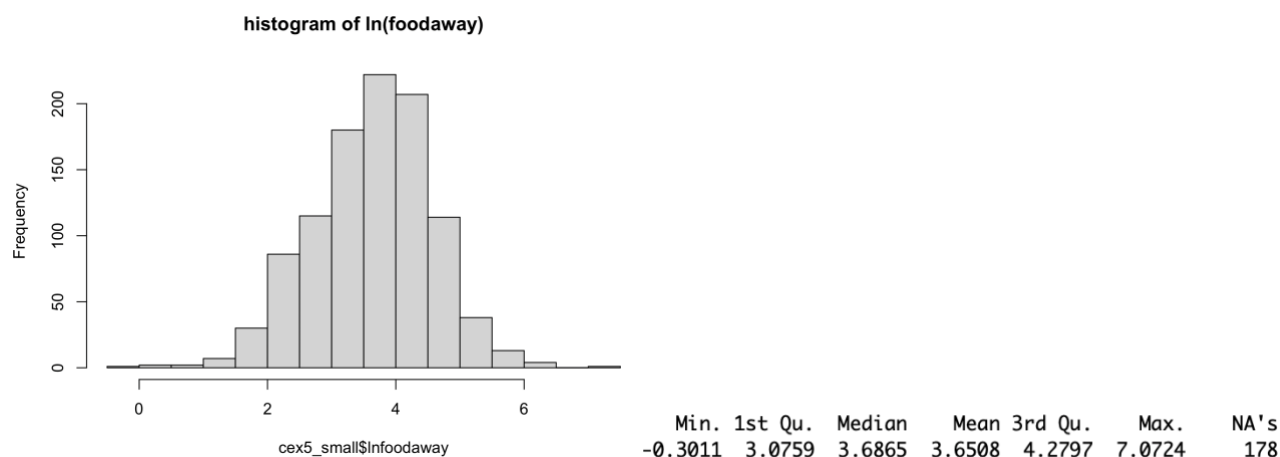
College=1

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	14.44	36.11	48.60	68.67	416.11

No advanced & college

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	9.63	26.02	39.01	52.65	437.78

(c.)



因為 $\ln(0)$ 無意義，所以相較原始會少了 178 個觀察值。

(d.)

```
Call:
lm(formula = lnfoodaway ~ income, data = cex5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-3.6547 -0.5777  0.0530  0.5937  2.7000

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.1293004   0.0565503   55.34  <2e-16 ***
income       0.0069017   0.0006546   10.54  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

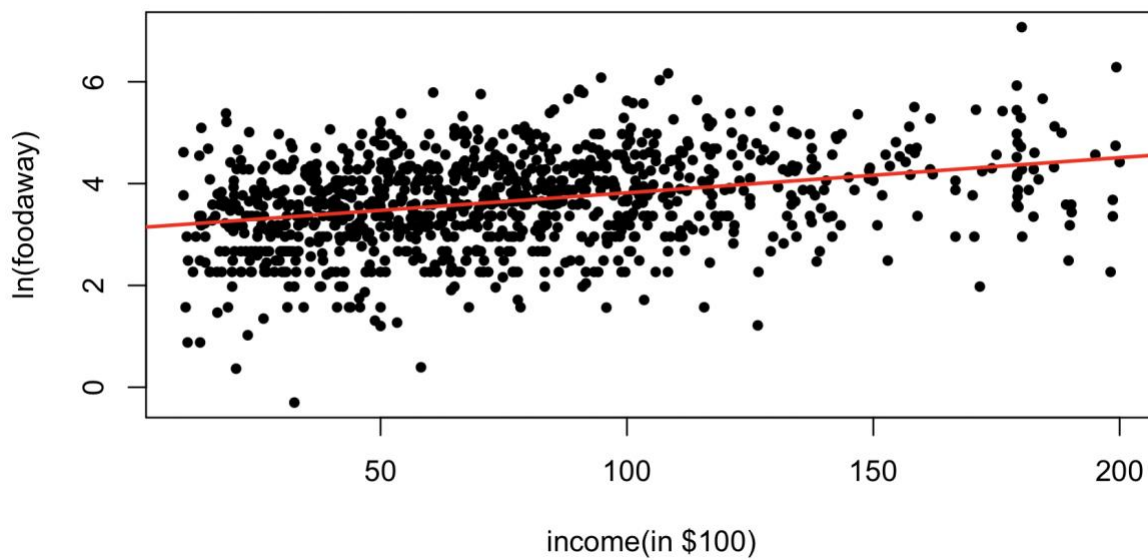
Residual standard error: 0.8761 on 1020 degrees of freedom
(178 observations deleted due to missingness)
Multiple R-squared:  0.09826, Adjusted R-squared:  0.09738
F-statistic: 111.1 on 1 and 1020 DF, p-value: < 2.2e-16
```

$$\widehat{\ln(\text{foodaway})}_i = 3.1293 + 0.0069 * \text{income}_i$$

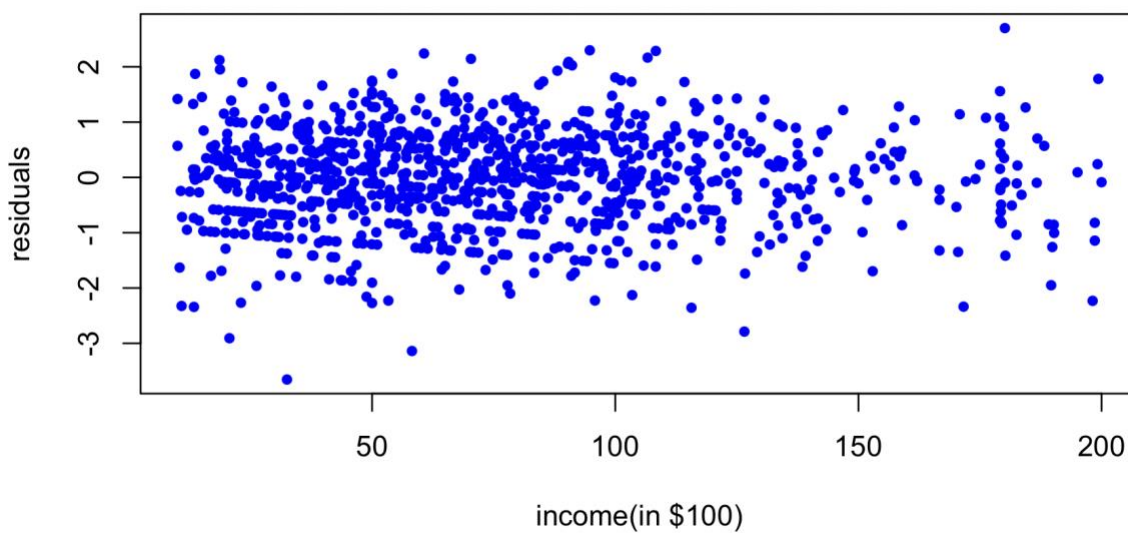
因為應變數進行對數轉換，所以該斜率的意涵便為每增加 1 單位 income，foodaway 會增加約

0.69%。

(e.)



(f.)



看起來 residual 的點是隨機分佈，而且不論收入高低其變異（散佈）程度並無太大差異【應當符合同質變異性 (Homoskedasticity)】。而在高收入的樣本，取得的樣本數相對中低收入而言

更少【不過社會當中中低收入的人本就佔大多數，該取樣應屬合理】。